

Predictive N_T — Seven-Component cCOSMO (with H2O)

PERSISTED QUALIFICATION STATUS — PRE-PILOT MULTISTAGE PREDICTIVE MODEL

Executive summary

Project 236 completed 10 simultaneous counter-current matrix trials. Predictive N_T is assigned only from a fully accepted physical multistage trial.

Project reference	236
Job status	COMPLETED — report generated automatically
N_T tested	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
Trials completed	10 / 10
Accepted Predictive N_T	4
Established theoretical stages	4
Pilot validated	No
Calibration required	No
Release eligible	No

ENGINEERING RESULT

Preferred engineering candidate: N_T = 4

Selection is reproducible: choose the smallest stage count whose complete simultaneous solution passes closure, positive-flow, phase, post-split TPD, branch-reproduction, and Stage-1 target gates. Experimental wet LLE is not an execution gate.

Allocation: 1237.55 feed-basis ppm; removal 64.641%; historical target PASS — BASIS MISMATCH

Selected trial versus frozen targets

Target (mass basis)	Result	Limit	Margin	Saved status
RRBO recovery %	87	80	7	PASS
Maximum NMP in raffinate (wt%)	8.68	20	11.32	PASS
Raffinate sulfur target (ppm)*	1237.55	2500	Mismatch	PASS
Minimum saturates (wt%)	92.8	90	2.8	PASS
Maximum polar aromatics (wt%)	0.38	0.5	0.12	PASS
Maximum total aromatics (wt%)	6.81	7	0.19	PASS

*BASIS MISMATCH: result is feed-basis retained sulfur; target is intended raffinate concentration. Historical PASS is preserved, NOT demonstrated product-concentration compliance.

Contents and reading guide

Frozen acceptance is preserved. Reporting-derived component removal does not re-evaluate qualification. Green comparison rows identify the saved selected trial; unaccepted trials remain diagnostic.

Main report and appendix groups are indexed below; PDF bookmarks provide every trial page.

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1. Frozen Design Basis

OWNER-CONTROLLED / SAVED STAGE-1 INPUT AUTHORITY

Project reference	236
RRBO grade	SN300
Operating temperature	40 °C
Operating pressure	2.0 bar(g)
Phase configuration	rrbo-continuous-nmp-dispersed
Solvent/oil mass ratio	0.6
Maximum stages	10
N_T tested	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
SAT identity	n-dodecane
MONO identity	n-propylbenzene

Feed composition — mass basis

SAT	MONO	DI	POLY	PA	NMP	H2O
85 wt%	7 wt%	4 wt%	2 wt%	2 wt%	0 wt%	0 wt%

Saved product targets

Maximum total aromatics	7 wt% NMP-free
Maximum polar aromatics	0.5 wt% NMP-free
Minimum saturates	90 wt% NMP-free
Minimum RRBO recovery	80 wt%
Maximum NMP in raffinate	20 wt% full stream
Raffinate sulfur target	2500 ppm

1.2 Frozen solvent and stream basis

Saved Stage-1 conditions and actual frozen boundary construction

Fresh-solvent NMP purity	99.5 wt%
Fresh-solvent water	0.5 wt%
S/O construction	FIXED_TOTAL_WET_SOLVENT_MASS
Total wet solvent / oil mass	0.6
Dry NMP / oil mass	0.597
Water / oil mass	0.003
Physical design feed rate	4000 L/h
Frozen boundary oil mass	100
Frozen boundary wet-solvent mass	60

Boundary quantities use the persisted normalized stream basis, not plant molar flow. Component mass divided by component moles gives the frozen molecular weight (g/mol on a gram/mol normalization). RRBO recovery and product composition exclude NMP and H₂O; total aromatics = MONO + DI + POLY, with PA reported separately. NMP and H₂O contents use the full raffinate stream. S/O is total wet-solvent mass / oil-feed mass when the frozen construction says FIXED_TOTAL_WET_SOLVENT_MASS.

RRBO / NMP density	869 / 1015 kg/m ³
RRBO / NMP dynamic viscosity	59.8 / 1.42 cP
Saved interfacial tension	11 mN/m
Saved basis notes	No notes recorded

1.3 Frozen thermodynamic basis

NIST COSMO-SAC-2010 / cCOSMO implementation and job-specific amendments

Base activity model	COSMO-SAC-2010
Actual model identity	NATIVE_SEVEN_COMPONENT_CCOSMO_2010_PLUS_ADDITIVE_REDlich_KISTER
Engine version / contract	7C-1.6.0
Native Gibbs contribution retained	true
Inherited residual applied	false
Additive interaction form	$\text{sum_}[\text{pairs}, k=0..2] [a+b*(T_{\text{ref}}/T-1)]*x_i*x_j*(x_i-x_j)^k$
Interaction pairs / parameters	8 / 48

cCOSMO identifies the implementation, not a different molecular basis. For this native-plus-RK contract the NIST COSMO-SAC-2010 contribution is retained with the persisted additive Redlich-Kister interaction model. This is not an unamended COSMO-SAC prediction. Model and parameter digests, molecular generation routes and sigma-profile provenance follow in the provenance appendix. Missing artifacts are never replaced with current-model values.

1.4 Seven-component molecular identities

Identities from hash-verified frozen generation manifests; unavailable identities remain explicit

Family	Molecule / surrogate	CAS	MW g/mol*
SAT	N-DODECANE	112-40-3	170.3348
MONO	N-PROPYLBENZENE	103-65-1	120.194
DI	1-METHYLNAPHTHALENE	90-12-0	142.1971
POLY	PYRENE	129-00-0	202.2506
PA	BIS-P-CUMYLPHENYL-AMINE	10081-67-1	405.58
NMP	N-METHYL-2-PYRROLIDONE	872-50-4	99.1311
H2O	WATER	7732-18-5	18.01528

*Reporting reconstruction from persisted boundary component mass / component moles, not a replacement molecular-weight database. PA is a non-sulfur-bearing surrogate. H2O is explicitly included in equilibrium.

1.5 Frozen Design Basis — Sulfur Allocation

OWNER-CONTROLLED MASS ALLOCATION — saved with the completed Stage-1 snapshot

Component	Feed wt%	Sulfur allocation %	Feed-basis sulfur contribution ppm
SAT	85	0	0
MONO	7	5	175
DI	4	25	875
POLY	2	35	1225
PA	2	35	1225
Feed sulfur		3500 ppm	
Raffinate sulfur target		2500 ppm	
Allocation total		100 %	

This allocation is an owner-controlled mass-allocation basis. It is not a hidden factor, does not make sulfur a thermodynamic component, and does not alter the seven-component equilibrium solution.

2. Overall N_T Engineering Comparison

Primary process comparison. Product composition is wt% on an NMP-free hydrocarbon basis; NMP is wt% of total raffinate.

N_T	Closure	Recovery %	NMP wt%	SAT wt%	Arom wt%	PA wt%	Sulfur ppm*	Targets	Saved acceptance
1	CLOSED	91.32	11.96	89.57	9.33	1.1	2099.6	FAIL	NOT ACCEPTED
2	CLOSED	88.81	9.98	91.41	7.87	0.71	1605.5	FAIL	NOT ACCEPTED
3	CLOSED	87.63	9.12	92.31	7.18	0.51	1368.4	FAIL	NOT ACCEPTED
4	CLOSED	87	8.68	92.8	6.81	0.38	1237.5	PASS	ACCEPTED
5	CLOSED	86.64	8.45	93.09	6.61	0.3	1159	PASS	ACCEPTED
6	CLOSED	86.43	8.31	93.26	6.5	0.24	1108.8	PASS	ACCEPTED
7	CLOSED	86.3	8.22	93.37	6.43	0.2	1075.2	PASS	ACCEPTED
8	CLOSED	86.22	8.17	93.44	6.4	0.17	1051.7	PASS	ACCEPTED
9	CLOSED	86.16	8.14	93.48	6.38	0.14	N/C	PASS	NOT ACCEPTED
10	CLOSED	86.13	8.12	93.52	6.37	0.12	1021.9	PASS	ACCEPTED

*Sulfur BASIS MISMATCH: saved result is feed-basis retained sulfur; intended target is raffinate concentration. Historical target PASS is NOT demonstrated product-concentration compliance. All saved statuses/selection remain unchanged. Selected trial is highlighted.

3. Component Extraction Performance

Engineering trade-off between RRBO recovery and aromatic/polar-component removal

N_T	Solver Status	RRBO Recovery %	SAT Loss %	MONO Removal %	DI Removal %	POLY Removal %	PA Removal %
1	CLOSED	91.321	3.766	34.812	33.223	35.802	49.812
2	CLOSED	88.806	4.494	46.855	44.144	48.085	68.342
3	CLOSED	87.626	4.835	52.501	48.96	53.809	77.729
4	CLOSED	86.999	5.014	55.49	51.311	56.819	83.293
5	CLOSED	86.642	5.114	57.175	52.504	58.5	86.933
6	CLOSED	86.43	5.171	58.159	53.112	59.468	89.484
7	CLOSED	86.3	5.204	58.745	53.414	60.033	91.365
8	CLOSED	86.217	5.224	59.094	53.555	60.362	92.806
9	CLOSED	86.164	5.236	59.302	53.61	60.549	93.941
10	CLOSED	86.13	5.242	59.423	53.621	60.651	94.856

Reporting-derived where metrics were absent: removal/loss (%) = 100*(oil-feed component moles - final-raffinate component moles)/oil-feed component moles. The same component MW cancels, so component-relative mass removal is identical. No equilibrium rerun. Recovery uses persisted hydrocarbon mass recovery (or boundary component masses if absent). SAT = loss; NMP removal = not applicable. Unaccepted trials remain diagnostic.

4. Governed Sulfur Post-Processing

POST-PROCESSING MASS ALLOCATION — DOES NOT MODIFY THERMODYNAMICS OR EQUILIBRIUM

N_T	Closure	SAT ppm	MONO ppm	DI ppm	POLY ppm	PA ppm	Feed-basis ppm	Allocated removal %	Raffinate target ppm*	Historical target*
1	CLOSED	0	114.08	584.29	786.43	614.8	2099.61	40.011	2500	PASS
2	CLOSED	0	93	488.74	635.96	387.82	1605.52	54.128	2500	PASS
3	CLOSED	0	83.12	446.6	565.85	272.82	1368.39	60.903	2500	PASS
4	CLOSED	0	77.89	426.03	528.97	204.66	1237.55	64.641	2500	PASS
5	CLOSED	0	74.94	415.59	508.37	160.07	1158.98	66.886	2500	PASS
6	CLOSED	0	73.22	410.27	496.51	128.82	1108.82	68.319	2500	PASS
7	CLOSED	0	72.2	407.62	489.59	105.78	1075.19	69.28	2500	PASS
8	CLOSED	0	71.58	406.39	485.57	88.13	1051.68	69.952	2500	PASS
9	CLOSED	—	—	—	—	—	Not calculable	—	2500	NOT_CALCULABLE
10	CLOSED	0	71.01	405.82	482.03	63.02	1021.87	70.804	2500	PASS

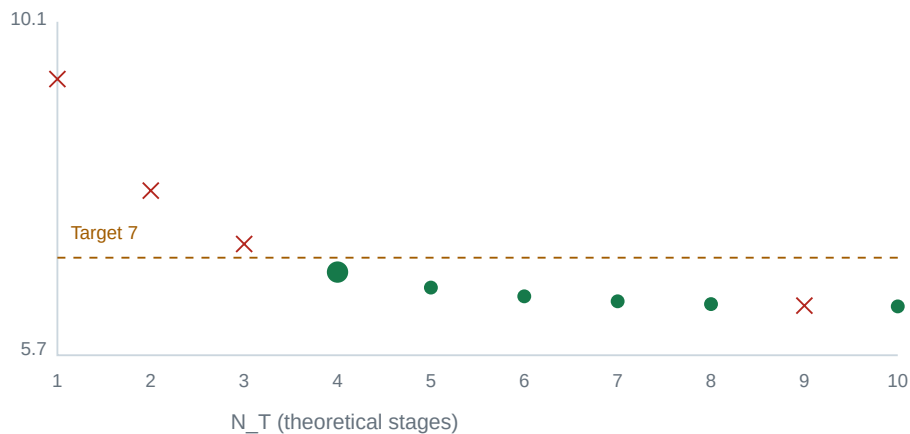
Frozen allocation basis: $S_{\text{retained}} = S_{\text{feed}} * \sum(\text{allocation}_i * m_{i,R} / m_{i,F})$. Allocated sulfur mass removal (%) = $100 * (1 - S_{\text{retained}} / S_{\text{feed}})$. Feed-basis ppm is NOT recovery-normalized raffinate concentration; the frozen model does not divide by RRBO recovery. Existing values and target decisions are preserved, not corrected scientifically.

*BASIS MISMATCH: the original target is intended raffinate sulfur concentration, but the saved comparison uses feed-basis retained sulfur. Historical PASS is NOT demonstrated product-concentration compliance. Concentration reduction = $100 * (1 - C_R / C_F)$ requires consistent concentration bases. Direct thermodynamic sulfur prediction remains NOT CALCULABLE. Scientific review requires separate authorization; no result or target decision is altered.

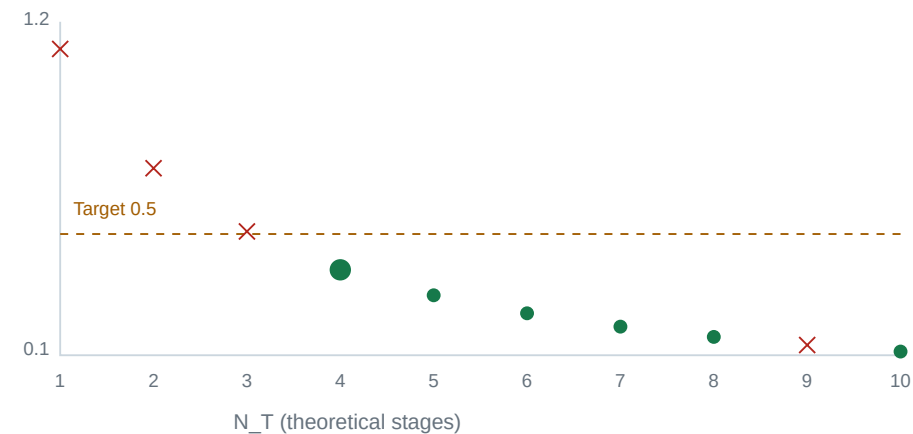
Engineering trends versus frozen targets

Filled green = accepted; red cross = not accepted. Points never confer scientific qualification.

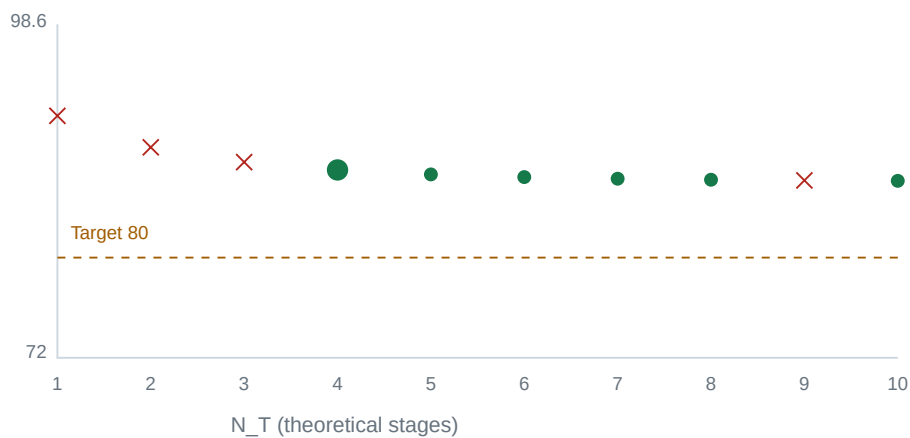
Total aromatics (wt%, hydrocarbon basis)



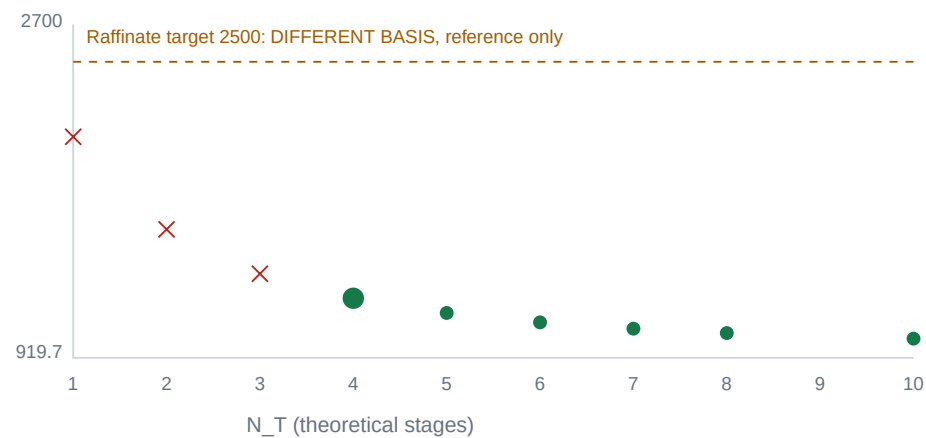
Polar aromatics (wt%, hydrocarbon basis)



RRBO mass recovery (%)



Allocated retained sulfur (feed-basis ppm)



5. Engineering Stage Selection

Preferred engineering candidate, accepted Predictive N_T, and established theoretical stages are separate states

Preferred engineering/process-sensitivity candidate	N_T = 4
Accepted Predictive N_T	4
Established theoretical stages	4

SELECTION GOVERNANCE

The persisted deterministic rule selects the smallest fully accepted simultaneous trial. Collapsed stages are NO_PHYSICAL_LLE and remain unselected; numerical or target failures cannot become Predictive N_T through presentation logic.

Immediately preceding trial N_T=3: Maximum polar aromatics (wt%): 0.5083 versus 0.5; Maximum total aromatics (wt%): 7.1787 versus 7. Selection shown above is persisted, not recomputed by this report.

6. Scientific Qualification Summary

Exact persisted evidence; PASS is never inferred from the absence of a blocker

Seven-component molecular basis incl. H2O	PERSISTED
Stage-1 authority	PERSISTED
Combined endpoint closure	PASS
Multistart: both endpoints closed	PASS
Global TPD stability	Per-stage TPD and explicit MONO-rich evidence in trial appendices
Sulfur thermodynamic prediction	NOT CALCULABLE
Pilot validation	NOT VALIDATED
Calibration requirement	NOT REQUIRED
Release eligibility	NOT ELIGIBLE
Accepted Predictive N_T	4

Main-report blocker summary

No exact blocker-code array is recorded. This is NOT evidence of acceptance. Appendix C lists persisted numerical, target and stage dispositions, including rejected trials.

Evidence completeness is separate from saved acceptance. In supported 7C matrix contracts, closure and termination status are combined endpoint fields. CLOSED establishes both endpoints closed; UNCLOSED does not identify the failing endpoint. Individual optimizer terminations are not recorded. Native residuals and branch differences are preserved; no gate or selected N_T changes.

Appendix A.1.1 — N_T=1 Numerical Diagnostics

NOT ACCEPTED (persisted) | numerical acceptance PASS | maximum component balance residual 2.880e-16

Combined endpoint status CONVERGED | Closure CLOSED | Scaled residual 1.757e-14 | Branch EVALUATED (1.239e-15)

Exact blocker codes: Not recorded; see persisted disposition in Appendix C

Targets: hydrocarbon mass basis; NMP full-stream; sulfur feed-basis result vs raffinate target: BASIS MISMATCH

Target	Status	Limit	Calculated
Minimum RRBO recovery (%)	PASS	80	91.3209
Maximum NMP in raffinate (wt%)	PASS	20	11.9561
Minimum saturates (wt%)	FAIL	90	89.5731
Maximum total aromatics (wt%)	FAIL	7	9.3277
Maximum polar aromatics (wt%)	FAIL	0.5	1.0992
Raffinate sulfur target (ppm)	PASS	2500	2099.6067

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Removal / loss	3.7659	34.8122	33.2234	35.8017	49.812	NOT APPLICABLE	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP	H2O
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0	0.0000e+0
fresh Wet Solvent	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	6.0223e-1	1.6653e-2
final Raffinate	4.8022e-1	3.7965e-2	1.8784e-2	6.3484e-3	2.4749e-3	1.2535e-1	1.0115e-2
final Extract	1.8793e-2	2.0274e-2	9.3457e-3	3.5403e-3	2.4563e-3	4.7689e-1	6.5370e-3

Overall seven-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Residual	8.327e-17	-1.041e-17	8.674e-18	-6.072e-18	1.301e-18	1.110e-16	-2.880e-16

Appendix A.1.2 — Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
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1	2.880e-16	PASS	1.774e-14	1.660e-3	1.698e-3	0.000e+0	0.000e+0
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Persisted refinement / explicit MONO-rich acceptance (not inferred from the minimum)

Stage	R refinements	E refinements	R MONO-rich	E MONO-rich
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1	PASS	PASS	PASS	PASS
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Appendix A.1.3 — Seven-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP	H2O
1 R	0.704907	0.055727	0.027573	0.009319	0.003633	0.183993	0.014848
1 E	0.034941	0.037696	0.017377	0.006583	0.004567	0.886681	0.012154

Appendix A.2.1 — N_T=2 Numerical Diagnostics

NOT ACCEPTED (persisted) | numerical acceptance PASS | maximum component balance residual 2.173e-15

Combined endpoint status CONVERGED | Closure CLOSED | Scaled residual 1.785e-13 | Branch EVALUATED (7.940e-14)

Exact blocker codes: Not recorded; see persisted disposition in Appendix C

Targets: hydrocarbon mass basis; NMP full-stream; sulfur feed-basis result vs raffinate target: BASIS MISMATCH

Target	Status	Limit	Calculated
Minimum RRBO recovery (%)	PASS	80	88.8064
Maximum NMP in raffinate (wt%)	PASS	20	9.9802
Minimum saturates (wt%)	PASS	90	91.4129
Maximum total aromatics (wt%)	FAIL	7	7.8741
Maximum polar aromatics (wt%)	FAIL	0.5	0.713
Raffinate sulfur target (ppm)	PASS	2500	1605.5236

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Removal / loss	4.4936	46.8547	44.1438	48.0848	68.3415	NOT APPLICABLE	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP	H2O
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0	0.0000e+0
fresh Wet Solvent	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	6.0223e-1	1.6653e-2
final Raffinate	4.7659e-1	3.0951e-2	1.5712e-2	5.1338e-3	1.5611e-3	9.9563e-2	1.2078e-2
final Extract	2.2424e-2	2.7288e-2	1.2418e-2	4.7550e-3	3.3701e-3	5.0267e-1	4.5748e-3

Overall seven-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Residual	3.469e-18	3.469e-17	5.204e-17	6.332e-17	2.173e-15	0.000e+0	-5.204e-17

Appendix A.2.2 — Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
1	1.314e-15	PASS	6.631e-14	1.215e-3	1.205e-3	0.000e+0	0.000e+0
2	8.587e-16	PASS	1.785e-13	1.691e-3	1.819e-3	0.000e+0	0.000e+0

Persisted refinement / explicit MONO-rich acceptance (not inferred from the minimum)

Stage	R refinements	E refinements	R MONO-rich	E MONO-rich
1	PASS	PASS	PASS	PASS
2	PASS	PASS	PASS	PASS

Appendix A.2.3 — Seven-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP	H2O
1 R	0.623122	0.070797	0.034906	0.012036	0.00509	0.245645	0.008404
1 E	0.038829	0.047252	0.021502	0.008234	0.005836	0.870426	0.007922
2 R	0.742828	0.048241	0.02449	0.008002	0.002433	0.155181	0.018825
2 E	0.033185	0.033364	0.015873	0.005836	0.003246	0.894013	0.014483

Appendix A.3.1 — N_T=3 Numerical Diagnostics

NOT ACCEPTED (persisted) | numerical acceptance PASS | maximum component balance residual 2.947e-13

Combined endpoint status CONVERGED | Closure CLOSED | Scaled residual 1.609e-13 | Branch EVALUATED (3.732e-13)

Exact blocker codes: Not recorded; see persisted disposition in Appendix C

Targets: hydrocarbon mass basis; NMP full-stream; sulfur feed-basis result vs raffinate target: BASIS MISMATCH

Target	Status	Limit	Calculated
Minimum RRBO recovery (%)	PASS	80	87.6262
Maximum NMP in raffinate (wt%)	PASS	20	9.1169
Minimum saturates (wt%)	PASS	90	92.313
Maximum total aromatics (wt%)	FAIL	7	7.1787
Maximum polar aromatics (wt%)	FAIL	0.5	0.5083
Raffinate sulfur target (ppm)	PASS	2500	1368.3861

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Removal / loss	4.8348	52.5006	48.9602	53.8085	77.7291	NOT APPLICABLE	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP	H2O
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0	0.0000e+0
fresh Wet Solvent	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	6.0223e-1	1.6653e-2
final Raffinate	4.7489e-1	2.7663e-2	1.4357e-2	4.5677e-3	1.0982e-3	8.8912e-2	1.3151e-2
final Extract	2.4127e-2	3.0576e-2	1.3772e-2	5.3210e-3	3.8330e-3	5.1332e-1	3.5015e-3

Overall seven-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Residual	7.633e-16	5.631e-15	7.367e-15	8.310e-15	2.947e-13	-2.331e-15	-8.014e-15

Appendix A.3.2 — Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
1	6.172e-14	PASS	6.506e-14	9.215e-4	9.217e-4	-7.175e-15	0.000e+0
2	1.065e-13	PASS	1.609e-13	1.450e-3	1.471e-3	0.000e+0	0.000e+0
3	1.265e-13	PASS	9.268e-14	1.461e-3	1.726e-3	0.000e+0	1.371e-15

Persisted refinement / explicit MONO-rich acceptance (not inferred from the minimum)

Stage	R refinements	E refinements	R MONO-rich	E MONO-rich
1	PASS	PASS	PASS	PASS
2	PASS	PASS	PASS	PASS
3	PASS	PASS	PASS	PASS

Appendix A.3.3 — Seven-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP	H2O
1 R	0.594471	0.076681	0.037492	0.013092	0.005755	0.266533	0.005975
1 E	0.040587	0.051436	0.023168	0.008951	0.006448	0.86352	0.00589
2 R	0.64647	0.066607	0.033606	0.0113	0.003821	0.227055	0.01114
2 E	0.03762	0.04453	0.020789	0.007786	0.004497	0.874626	0.010151
3 R	0.760262	0.044287	0.022985	0.007313	0.001758	0.142341	0.021054
3 E	0.03226	0.031046	0.015142	0.00543	0.002418	0.897916	0.015788

Appendix A.4.1 — N_T=4 Numerical Diagnostics

ACCEPTED (persisted) | numerical acceptance PASS | maximum component balance residual 1.041e-17

Combined endpoint status CONVERGED | Closure CLOSED | Scaled residual 3.197e-14 | Branch EVALUATED (2.185e-15)

Exact blocker codes: Not recorded; see persisted disposition in Appendix C

Targets: hydrocarbon mass basis; NMP full-stream; sulfur feed-basis result vs raffinate target: BASIS MISMATCH

Target	Status	Limit	Calculated
Minimum RRBO recovery (%)	PASS	80	86.9991
Maximum NMP in raffinate (wt%)	PASS	20	8.6838
Minimum saturates (wt%)	PASS	90	92.8034
Maximum total aromatics (wt%)	PASS	7	6.8126
Maximum polar aromatics (wt%)	PASS	0.5	0.3841
Raffinate sulfur target (ppm)	PASS	2500	1237.5478

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Removal / loss	5.014	55.4898	51.3114	56.8188	83.293	NOT APPLICABLE	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP	H2O
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0	0.0000e+0
fresh Wet Solvent	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	6.0223e-1	1.6653e-2
final Raffinate	4.7400e-1	2.5922e-2	1.3696e-2	4.2701e-3	8.2386e-4	8.3697e-2	1.3810e-2
final Extract	2.5021e-2	3.2317e-2	1.4434e-2	5.6187e-3	4.1074e-3	5.1854e-1	2.8423e-3

Overall seven-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Residual	1.041e-17	0.000e+0	-1.735e-18	-2.602e-18	8.674e-19	0.000e+0	-6.505e-18

Appendix A.4.2 — Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
1	4.510e-17	PASS	1.737e-14	7.412e-4	7.487e-4	0.000e+0	0.000e+0
2	1.110e-16	PASS	1.110e-14	1.191e-3	1.198e-3	0.000e+0	0.000e+0
3	1.110e-16	PASS	3.197e-14	1.515e-3	1.588e-3	0.000e+0	0.000e+0
4	5.204e-17	PASS	1.865e-14	1.195e-3	1.523e-3	0.000e+0	0.000e+0

Persisted refinement / explicit MONO-rich acceptance (not inferred from the minimum)

Stage	R refinements	E refinements	R MONO-rich	E MONO-rich
1	PASS	PASS	PASS	PASS
2	PASS	PASS	PASS	PASS
3	PASS	PASS	PASS	PASS
4	PASS	PASS	PASS	PASS

Appendix A.4.3 — Seven-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP	H2O
1 R	0.581148	0.079616	0.038623	0.013614	0.006139	0.276169	0.00469
1 E	0.041502	0.053604	0.023942	0.00932	0.006813	0.860104	0.004715
2 R	0.607439	0.074391	0.03706	0.01271	0.004561	0.255689	0.008149
2 E	0.039823	0.049866	0.022916	0.008712	0.005182	0.865589	0.007912
3 R	0.660363	0.064205	0.032994	0.010879	0.002969	0.215684	0.012905
3 E	0.03696	0.043061	0.020514	0.007548	0.003558	0.876828	0.011532
4 R	0.769206	0.042067	0.022226	0.006929	0.001337	0.135823	0.022411
4 E	0.03174	0.029727	0.014781	0.005199	0.00187	0.900067	0.016615

Appendix A.5.1 — N_T=5 Numerical Diagnostics

ACCEPTED (persisted) | numerical acceptance PASS | maximum component balance residual 2.222e-12

Combined endpoint status CONVERGED | Closure CLOSED | Scaled residual 4.971e-13 | Branch EVALUATED (2.278e-12)

Exact blocker codes: Not recorded; see persisted disposition in Appendix C

Targets: hydrocarbon mass basis; NMP full-stream; sulfur feed-basis result vs raffinate target: BASIS MISMATCH

Target	Status	Limit	Calculated
Minimum RRBO recovery (%)	PASS	80	86.6423
Maximum NMP in raffinate (wt%)	PASS	20	8.4461
Minimum saturates (wt%)	PASS	90	93.0878
Maximum total aromatics (wt%)	PASS	7	6.6106
Maximum polar aromatics (wt%)	PASS	0.5	0.3016
Raffinate sulfur target (ppm)	PASS	2500	1158.9809

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Removal / loss	5.1136	57.1752	52.504	58.5001	86.9328	NOT APPLICABLE	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP	H2O
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0	0.0000e+0
fresh Wet Solvent	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	6.0223e-1	1.6653e-2
final Raffinate	4.7350e-1	2.4941e-2	1.3361e-2	4.1038e-3	6.4437e-4	8.0870e-2	1.4247e-2
final Extract	2.5518e-2	3.3298e-2	1.4769e-2	5.7849e-3	4.2868e-3	5.2136e-1	2.4052e-3

Overall seven-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Residual	5.969e-14	-1.086e-13	-1.303e-13	-1.343e-13	-1.753e-13	2.554e-15	-2.222e-12

Appendix A.5.2 — Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
1	5.350e-13	PASS	5.684e-14	6.232e-4	6.344e-4	0.000e+0	4.278e-15
2	6.060e-13	PASS	4.263e-14	1.002e-3	1.014e-3	0.000e+0	0.000e+0
3	5.175e-13	PASS	1.394e-13	1.335e-3	1.360e-3	0.000e+0	0.000e+0
4	3.673e-13	PASS	5.046e-14	1.482e-3	1.612e-3	0.000e+0	0.000e+0
5	1.965e-13	PASS	1.737e-14	9.750e-4	1.300e-3	0.000e+0	0.000e+0

Persisted refinement / explicit MONO-rich acceptance (not inferred from the minimum)

Stage	R refinements	E refinements	R MONO-rich	E MONO-rich
1	PASS	PASS	PASS	PASS
2	PASS	PASS	PASS	PASS
3	PASS	PASS	PASS	PASS
4	PASS	PASS	PASS	PASS
5	PASS	PASS	PASS	PASS

Appendix A.5.3 — Seven-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP	H2O
1 R	0.574067	0.081238	0.03915	0.013899	0.006391	0.281353	0.003901
1 E	0.04201	0.054819	0.024315	0.009524	0.007057	0.858316	0.00396
2 R	0.589212	0.078371	0.038572	0.013424	0.005015	0.268837	0.006569
2 E	0.041024	0.05275	0.023924	0.009208	0.00562	0.86092	0.006554
3 R	0.615443	0.073083	0.036938	0.012498	0.003687	0.248707	0.009644
3 E	0.039386	0.049017	0.022875	0.00859	0.004232	0.866619	0.009282
4 R	0.66873	0.062791	0.032725	0.010635	0.002369	0.208638	0.014111
4 E	0.036581	0.042237	0.020433	0.007418	0.002874	0.877984	0.012473
5 R	0.774115	0.040775	0.021843	0.006709	0.001053	0.132212	0.023293
5 E	0.031439	0.028957	0.01461	0.005065	0.001488	0.901261	0.01718

Appendix A.6.1 — N_T=6 Numerical Diagnostics

ACCEPTED (persisted) | numerical acceptance PASS | maximum component balance residual 1.616e-12

Combined endpoint status CONVERGED | Closure CLOSED | Scaled residual 3.067e-13 | Branch EVALUATED (1.612e-12)

Exact blocker codes: Not recorded; see persisted disposition in Appendix C

Targets: hydrocarbon mass basis; NMP full-stream; sulfur feed-basis result vs raffinate target: BASIS MISMATCH

Target	Status	Limit	Calculated
Minimum RRBO recovery (%)	PASS	80	86.4301
Maximum NMP in raffinate (wt%)	PASS	20	8.3079
Minimum saturates (wt%)	PASS	90	93.2601
Maximum total aromatics (wt%)	PASS	7	6.4966
Maximum polar aromatics (wt%)	PASS	0.5	0.2433
Raffinate sulfur target (ppm)	PASS	2500	1108.8238

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Removal / loss	5.1708	58.1594	53.1119	59.4684	89.4841	NOT APPLICABLE	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP	H2O
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0	0.0000e+0
fresh Wet Solvent	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	6.0223e-1	1.6653e-2
final Raffinate	4.7321e-1	2.4368e-2	1.3190e-2	4.0081e-3	5.1856e-4	7.9237e-2	1.4554e-2
final Extract	2.5803e-2	3.3872e-2	1.4940e-2	5.8807e-3	4.4126e-3	5.2300e-1	2.0982e-3

Overall seven-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Residual	4.433e-14	-8.339e-14	-9.952e-14	-1.019e-13	-8.963e-14	2.776e-15	-1.616e-12

Appendix A.6.2 — Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
1	3.097e-13	PASS	1.910e-13	5.433e-4	5.538e-4	2.415e-15	0.000e+0
2	3.739e-13	PASS	8.329e-14	8.702e-4	8.852e-4	0.000e+0	0.000e+0
3	3.485e-13	PASS	4.796e-14	1.168e-3	1.189e-3	-8.066e-15	-3.567e-15
4	2.812e-13	PASS	1.865e-14	1.400e-3	1.451e-3	-3.792e-15	0.000e+0
5	1.971e-13	PASS	3.370e-14	1.387e-3	1.565e-3	0.000e+0	0.000e+0
6	1.055e-13	PASS	5.773e-14	8.036e-4	1.099e-3	0.000e+0	0.000e+0

Persisted refinement / explicit MONO-rich acceptance (not inferred from the minimum)

Stage	R refinements	E refinements	R MONO-rich	E MONO-rich
1	PASS	PASS	PASS	PASS
2	PASS	PASS	PASS	PASS
3	PASS	PASS	PASS	PASS
4	PASS	PASS	PASS	PASS
5	PASS	PASS	PASS	PASS
6	PASS	PASS	PASS	PASS

Appendix A.6.3 — Seven-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP	H2O
1 R	0.570028	0.082183	0.039394	0.014063	0.00657	0.284391	0.003371
1 E	0.0423	0.055527	0.024492	0.00964	0.007234	0.857367	0.00344
2 R	0.579491	0.080611	0.039271	0.013821	0.005326	0.275895	0.005586
2 E	0.041708	0.054407	0.024414	0.009489	0.005927	0.858405	0.00565
3 R	0.594366	0.077684	0.038651	0.013327	0.004149	0.263888	0.007935
3 E	0.040739	0.052309	0.024	0.009159	0.004684	0.861224	0.007885
4 R	0.620485	0.072332	0.036956	0.01238	0.003029	0.24409	0.010728
4 E	0.039128	0.048558	0.022924	0.00853	0.003501	0.867082	0.010276
5 R	0.673855	0.061949	0.03263	0.010493	0.00193	0.204169	0.014974
5 E	0.036357	0.041771	0.020441	0.007348	0.00236	0.878566	0.013157
6 R	0.776921	0.040007	0.021655	0.00658	0.000851	0.130091	0.023895
6 E	0.031259	0.028502	0.014537	0.004988	0.00121	0.901916	0.017587

Appendix A.7.1 — N_T=7 Numerical Diagnostics

ACCEPTED (persisted) | numerical acceptance PASS | maximum component balance residual 9.257e-16

Combined endpoint status CONVERGED | Closure CLOSED | Scaled residual 1.315e-13 | Branch EVALUATED (1.816e-14)

Exact blocker codes: Not recorded; see persisted disposition in Appendix C

Targets: hydrocarbon mass basis; NMP full-stream; sulfur feed-basis result vs raffinate target: BASIS MISMATCH

Target	Status	Limit	Calculated
Minimum RRBO recovery (%)	PASS	80	86.2997
Maximum NMP in raffinate (wt%)	PASS	20	8.2241
Minimum saturates (wt%)	PASS	90	93.3681
Maximum total aromatics (wt%)	PASS	7	6.4318
Maximum polar aromatics (wt%)	PASS	0.5	0.2001
Raffinate sulfur target (ppm)	PASS	2500	1075.1919

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Removal / loss	5.2043	58.7445	53.4143	60.0332	91.3651	NOT APPLICABLE	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP	H2O
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0	0.0000e+0
fresh Wet Solvent	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	6.0223e-1	1.6653e-2
final Raffinate	4.7305e-1	2.4027e-2	1.3105e-2	3.9522e-3	4.2580e-4	7.8252e-2	1.4780e-2
final Extract	2.5970e-2	3.4212e-2	1.5025e-2	5.9365e-3	4.5054e-3	5.2398e-1	1.8725e-3

Overall seven-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Residual	7.286e-17	-6.245e-17	-5.378e-17	-6.418e-17	-1.396e-16	-3.331e-16	-9.257e-16

Appendix A.7.2 — Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
1	1.511e-16	PASS	7.272e-14	4.845e-4	4.953e-4	5.612e-15	0.000e+0
2	2.030e-16	PASS	3.730e-14	7.751e-4	7.913e-4	0.000e+0	0.000e+0
3	2.220e-16	PASS	1.315e-13	1.041e-3	1.063e-3	0.000e+0	0.000e+0
4	1.579e-16	PASS	3.697e-14	1.270e-3	1.305e-3	0.000e+0	-9.358e-16
5	1.110e-16	PASS	2.315e-14	1.406e-3	1.487e-3	0.000e+0	0.000e+0
6	7.633e-17	PASS	2.220e-14	1.258e-3	1.466e-3	0.000e+0	0.000e+0
7	1.110e-16	PASS	3.464e-14	6.709e-4	9.303e-4	0.000e+0	0.000e+0

Persisted refinement / explicit MONO-rich acceptance (not inferred from the minimum)

Stage	R refinements	E refinements	R MONO-rich	E MONO-rich
1	PASS	PASS	PASS	PASS
2	PASS	PASS	PASS	PASS
3	PASS	PASS	PASS	PASS
4	PASS	PASS	PASS	PASS
5	PASS	PASS	PASS	PASS
6	PASS	PASS	PASS	PASS
7	PASS	PASS	PASS	PASS

Appendix A.7.3 — Seven-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP	H2O
1 R	0.567621	0.082749	0.039499	0.014159	0.006704	0.286275	0.002993
1 E	0.04247	0.055948	0.024571	0.009708	0.007368	0.856873	0.003062
2 R	0.573924	0.081933	0.039591	0.014051	0.005553	0.280032	0.004915
2 E	0.042108	0.05539	0.024647	0.009652	0.006154	0.857041	0.005008
3 R	0.583102	0.080301	0.039436	0.013792	0.004476	0.272026	0.006866
3 E	0.041523	0.054229	0.024545	0.009487	0.005011	0.858272	0.006933
4 R	0.597737	0.077319	0.038778	0.013283	0.00347	0.260445	0.00897
4 E	0.040567	0.052097	0.024109	0.009146	0.003938	0.861247	0.008896
5 R	0.623724	0.071909	0.037033	0.012319	0.00252	0.24095	0.011545
5 E	0.038974	0.048321	0.023007	0.008505	0.002928	0.867233	0.011032
6 R	0.677054	0.06145	0.032618	0.010412	0.001596	0.201256	0.015614
6 E	0.036223	0.041512	0.020485	0.007312	0.001964	0.878827	0.013677
7 R	0.778565	0.039545	0.021568	0.006505	0.000701	0.128791	0.024326
7 E	0.031151	0.028233	0.014515	0.004943	0.001	0.902265	0.017893

Appendix A.8.1 — N_T=8 Numerical Diagnostics

ACCEPTED (persisted) | numerical acceptance PASS | maximum component balance residual 2.220e-16

Combined endpoint status CONVERGED | Closure CLOSED | Scaled residual 7.550e-14 | Branch EVALUATED (4.498e-15)

Exact blocker codes: Not recorded; see persisted disposition in Appendix C

Targets: hydrocarbon mass basis; NMP full-stream; sulfur feed-basis result vs raffinate target: BASIS MISMATCH

Target	Status	Limit	Calculated
Minimum RRBO recovery (%)	PASS	80	86.2175
Maximum NMP in raffinate (wt%)	PASS	20	8.1717
Minimum saturates (wt%)	PASS	90	93.4377
Maximum total aromatics (wt%)	PASS	7	6.3954
Maximum polar aromatics (wt%)	PASS	0.5	0.1669
Raffinate sulfur target (ppm)	PASS	2500	1051.6796

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Removal / loss	5.224	59.0944	53.555	60.3616	92.8056	NOT APPLICABLE	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP	H2O
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0	0.0000e+0
fresh Wet Solvent	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	6.0223e-1	1.6653e-2
final Raffinate	4.7295e-1	2.3823e-2	1.3065e-2	3.9197e-3	3.5477e-4	7.7639e-2	1.4952e-2
final Extract	2.6069e-2	3.4416e-2	1.5065e-2	5.9690e-3	4.5764e-3	5.2459e-1	1.7004e-3

Overall seven-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Residual	-1.388e-17	1.388e-17	-8.674e-18	0.000e+0	-6.852e-17	-2.220e-16	-4.770e-17

Appendix A.8.2 — Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
1	1.388e-17	PASS	2.665e-14	4.402e-4	4.504e-4	-2.525e-15	0.000e+0
2	1.110e-16	PASS	5.773e-14	7.030e-4	7.199e-4	-2.861e-15	0.000e+0
3	1.110e-16	PASS	1.776e-14	9.443e-4	9.675e-4	0.000e+0	0.000e+0
4	1.110e-16	PASS	1.243e-14	1.158e-3	1.190e-3	-5.317e-15	0.000e+0
5	1.110e-16	PASS	2.842e-14	1.323e-3	1.376e-3	0.000e+0	0.000e+0
6	1.665e-16	PASS	7.550e-14	1.366e-3	1.478e-3	0.000e+0	0.000e+0
7	1.804e-16	PASS	2.132e-14	1.120e-3	1.336e-3	0.000e+0	0.000e+0
8	2.429e-17	PASS	5.418e-14	5.646e-4	7.906e-4	0.000e+0	0.000e+0

Persisted refinement / explicit MONO-rich acceptance (not inferred from the minimum)

Stage	R refinements	E refinements	R MONO-rich	E MONO-rich
1	PASS	PASS	PASS	PASS
2	PASS	PASS	PASS	PASS
3	PASS	PASS	PASS	PASS
4	PASS	PASS	PASS	PASS
5	PASS	PASS	PASS	PASS
6	PASS	PASS	PASS	PASS
7	PASS	PASS	PASS	PASS
8	PASS	PASS	PASS	PASS

Appendix A.8.3 — Seven-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP	H2O
1 R	0.566141	0.083092	0.039536	0.014216	0.006809	0.287496	0.002709
1 E	0.042569	0.0562	0.0246	0.009747	0.007473	0.856635	0.002777
2 R	0.570595	0.082732	0.039726	0.014187	0.005728	0.282604	0.004428
2 E	0.042343	0.055982	0.024749	0.009748	0.006331	0.856317	0.004529
3 R	0.576637	0.08186	0.039789	0.014065	0.004722	0.276799	0.006129
3 E	0.041986	0.055381	0.024802	0.00968	0.00526	0.856651	0.006241
4 R	0.585567	0.080176	0.039604	0.013792	0.00379	0.2692	0.00787
4 E	0.041408	0.054182	0.02468	0.009503	0.004262	0.858024	0.007942
5 R	0.599991	0.077143	0.038915	0.013268	0.002926	0.257975	0.00978
5 E	0.040463	0.052017	0.024222	0.009151	0.003335	0.861121	0.009692
6 R	0.625849	0.071681	0.037128	0.01229	0.002118	0.238753	0.012182
6 E	0.038882	0.048212	0.023097	0.0085	0.00247	0.867212	0.011627
7 R	0.679088	0.061159	0.032645	0.010367	0.001336	0.1993	0.016105
7 E	0.036142	0.041376	0.020541	0.007297	0.001651	0.878907	0.014086
8 R	0.779541	0.039267	0.021534	0.006461	0.000585	0.127968	0.024645
8 E	0.031084	0.028076	0.014517	0.004919	0.000837	0.902437	0.018131

Appendix A.9.1 — N_T=9 Numerical Diagnostics

NOT ACCEPTED (persisted) | numerical acceptance FAIL | maximum component balance residual 2.888e-13

Combined endpoint status CONVERGED | Closure CLOSED | Scaled residual 1.252e-13 | Branch EVALUATED (3.183e-13)

Exact blocker codes: Not recorded; see persisted disposition in Appendix C

Targets: hydrocarbon mass basis; NMP full-stream; sulfur feed-basis result vs raffinate target: BASIS MISMATCH

Target	Status	Limit	Calculated
Minimum RRBO recovery (%)	PASS	80	86.1645
Maximum NMP in raffinate (wt%)	PASS	20	8.1381
Minimum saturates (wt%)	PASS	90	93.4838
Maximum total aromatics (wt%)	PASS	7	6.3756
Maximum polar aromatics (wt%)	PASS	0.5	0.1406
Raffinate sulfur target (ppm)	NOT CALCULABLE	2500	—

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Removal / loss	5.2355	59.3023	53.6102	60.5488	93.9409	NOT APPLICABLE	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP	H2O
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0	0.0000e+0
fresh Wet Solvent	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	6.0223e-1	1.6653e-2
final Raffinate	4.7289e-1	2.3702e-2	1.3049e-2	3.9012e-3	2.9879e-4	7.7246e-2	1.5087e-2
final Extract	2.6126e-2	3.4537e-2	1.5081e-2	5.9875e-3	4.6324e-3	5.2499e-1	1.5653e-3

Overall seven-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Residual	-8.812e-16	-5.003e-15	-6.330e-15	-6.668e-15	-2.888e-13	2.331e-15	1.054e-14

Appendix A.9.2 — Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
1	7.615e-15	PASS	1.243e-14	4.053e-4	4.151e-4	0.000e+0	0.000e+0
2	1.332e-14	PASS	1.954e-14	6.479e-4	6.640e-4	0.000e+0	0.000e+0
3	1.934e-14	PASS	6.128e-14	8.705e-4	8.929e-4	0.000e+0	0.000e+0
4	2.573e-14	PASS	2.931e-14	1.069e-3	1.100e-3	0.000e+0	0.000e+0
5	3.233e-14	PASS	9.415e-14	1.235e-3	1.278e-3	0.000e+0	0.000e+0
6	3.941e-14	PASS	8.971e-14	1.336e-3	1.410e-3	-2.838e-15	0.000e+0
7	4.660e-14	PASS	8.260e-14	1.290e-3	1.428e-3	0.000e+0	0.000e+0
8	5.337e-14	FAIL	1.252e-13	9.858e-4	1.194e-3	1.195e-16	0.000e+0
9	5.106e-14	PASS	5.151e-14	4.793e-4	6.752e-4	0.000e+0	0.000e+0

Persisted refinement / explicit MONO-rich acceptance (not inferred from the minimum)

Stage	R refinements	E refinements	R MONO-rich	E MONO-rich
1	PASS	PASS	PASS	PASS
2	PASS	PASS	PASS	PASS
3	PASS	PASS	PASS	PASS
4	PASS	PASS	PASS	PASS
5	PASS	PASS	PASS	PASS
6	PASS	PASS	PASS	PASS
7	PASS	PASS	PASS	PASS
8	FAIL	PASS	PASS	PASS
9	PASS	PASS	PASS	PASS

Appendix A.9.3 — Seven-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP	H2O
1 R	0.565209	0.0833	0.03954	0.014249	0.006893	0.28832	0.00249
1 E	0.042626	0.056349	0.024605	0.009769	0.007558	0.85654	0.002554
2 R	0.568543	0.083221	0.03977	0.014269	0.005866	0.284275	0.004057
2 E	0.042482	0.056339	0.024784	0.009804	0.006472	0.855959	0.004159
3 R	0.572761	0.082808	0.039934	0.014227	0.004913	0.27977	0.005586
3 E	0.04226	0.05608	0.024912	0.009793	0.005457	0.855782	0.005716
4 R	0.578574	0.081886	0.039971	0.014092	0.004036	0.274334	0.007107
4 E	0.041907	0.055441	0.024946	0.009714	0.004514	0.856239	0.007238
5 R	0.587292	0.080156	0.039761	0.013807	0.003229	0.267077	0.008678
5 E	0.041337	0.054207	0.024805	0.009527	0.003644	0.857723	0.008757
6 R	0.601535	0.077079	0.039044	0.013271	0.002486	0.256153	0.010431
6 E	0.0404	0.052011	0.024329	0.009165	0.002842	0.860919	0.010335
7 R	0.627272	0.071571	0.037223	0.012281	0.001794	0.237169	0.012691
7 E	0.038827	0.048177	0.023182	0.008506	0.002099	0.867101	0.012108
8 R	0.680406	0.060995	0.032687	0.010344	0.001129	0.197947	0.016492
8 E	0.036094	0.041313	0.020598	0.007293	0.001398	0.878887	0.014417
9 R	0.780122	0.039101	0.021527	0.006436	0.000493	0.127431	0.024889
9 E	0.031044	0.027987	0.014529	0.004906	0.000706	0.902507	0.01832

Appendix A.10.1 — N_T=10 Numerical Diagnostics

ACCEPTED (persisted) | numerical acceptance PASS | maximum component balance residual 1.271e-14

Combined endpoint status CONVERGED | Closure CLOSED | Scaled residual 8.615e-14 | Branch EVALUATED (2.049e-14)

Exact blocker codes: Not recorded; see persisted disposition in Appendix C

Targets: hydrocarbon mass basis; NMP full-stream; sulfur feed-basis result vs raffinate target: BASIS MISMATCH

Target	Status	Limit	Calculated
Minimum RRBO recovery (%)	PASS	80	86.1296
Maximum NMP in raffinate (wt%)	PASS	20	8.116
Minimum saturates (wt%)	PASS	90	93.5151
Maximum total aromatics (wt%)	PASS	7	6.3654
Maximum polar aromatics (wt%)	PASS	0.5	0.1195
Raffinate sulfur target (ppm)	PASS	2500	1021.8737

Boundary-mole removal diagnostics — SAT loss and aromatic-family removal (%)

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Removal / loss	5.2422	59.4232	53.6211	60.6506	94.8556	NOT APPLICABLE	NOT APPLICABLE

Component moles	SAT	MONO	DI	POLY	PA	NMP	H2O
oil Feed	4.9902e-1	5.8239e-2	2.8130e-2	9.8887e-3	4.9312e-3	0.0000e+0	0.0000e+0
fresh Wet Solvent	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	0.0000e+0	6.0223e-1	1.6653e-2
final Raffinate	4.7286e-1	2.3632e-2	1.3046e-2	3.8912e-3	2.5368e-4	7.6988e-2	1.5196e-2
final Extract	2.6159e-2	3.4608e-2	1.5084e-2	5.9976e-3	4.6775e-3	5.2524e-1	1.4566e-3

Overall seven-component balance residual — component moles

Basis	SAT	MONO	DI	POLY	PA	NMP	H2O
Residual	9.714e-17	-1.943e-16	-2.394e-16	-2.663e-16	-1.271e-14	2.220e-16	4.705e-16

Appendix A.10.2 — Stage TPD and Stability Evidence

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

Stage	Local balance	Stage qualification	Isoactivity	R min eig	E min eig	R TPD min	E TPD min
1	2.784e-16	PASS	2.309e-14	3.776e-4	3.869e-4	0.000e+0	0.000e+0
2	4.892e-16	PASS	1.174e-14	6.035e-4	6.191e-4	0.000e+0	0.000e+0
3	7.321e-16	PASS	2.276e-14	8.115e-4	8.330e-4	2.237e-15	0.000e+0
4	9.897e-16	PASS	1.421e-14	9.978e-4	1.027e-3	0.000e+0	0.000e+0
5	1.288e-15	PASS	1.754e-14	1.157e-3	1.198e-3	-6.992e-16	0.000e+0
6	1.594e-15	PASS	6.306e-14	1.276e-3	1.334e-3	0.000e+0	0.000e+0
7	1.833e-15	PASS	2.076e-14	1.313e-3	1.408e-3	0.000e+0	0.000e+0
8	1.929e-15	PASS	6.484e-14	1.191e-3	1.344e-3	0.000e+0	0.000e+0
9	1.961e-15	PASS	8.615e-14	8.636e-4	1.057e-3	0.000e+0	0.000e+0
10	1.610e-15	PASS	4.086e-14	4.086e-4	5.792e-4	0.000e+0	0.000e+0

Persisted refinement / explicit MONO-rich acceptance (not inferred from the minimum)

Stage	R refinements	E refinements	R MONO-rich	E MONO-rich
1	PASS	PASS	PASS	PASS
2	PASS	PASS	PASS	PASS
3	PASS	PASS	PASS	PASS
4	PASS	PASS	PASS	PASS
5	PASS	PASS	PASS	PASS
6	PASS	PASS	PASS	PASS
7	PASS	PASS	PASS	PASS
8	PASS	PASS	PASS	PASS
9	PASS	PASS	PASS	PASS
10	PASS	PASS	PASS	PASS

Appendix A.10.3 — Seven-Component Stage Outlet Compositions

Stage numbering is from bottom to top: Stage 1 = RRBO-feed/bottom end; final stage = fresh-NMP/top end.

PHASE MOLE FRACTIONS — NOT PRODUCT MASS FRACTIONS

Stage / phase	SAT	MONO	DI	POLY	PA	NMP	H2O
1 R	0.564609	0.083424	0.03953	0.014268	0.006961	0.288894	0.002314
1 E	0.042658	0.056435	0.024597	0.00978	0.007628	0.856526	0.002375
2 R	0.567247	0.083518	0.039771	0.014316	0.005978	0.285404	0.003765
2 E	0.042563	0.056553	0.024787	0.009835	0.006587	0.85581	0.003865
3 R	0.570369	0.08339	0.039977	0.014324	0.005067	0.281703	0.005169
3 E	0.042423	0.056504	0.024948	0.00986	0.005615	0.855346	0.005303
4 R	0.57438	0.082932	0.040117	0.014271	0.00423	0.277529	0.006542
4 E	0.042205	0.056209	0.025058	0.00984	0.004716	0.855276	0.006697
5 R	0.579994	0.081966	0.040132	0.014124	0.003464	0.272407	0.007912
5 E	0.041857	0.055535	0.025075	0.009751	0.003888	0.855832	0.008062
6 R	0.588529	0.080196	0.0399	0.013829	0.002764	0.265439	0.009342
6 E	0.041293	0.054271	0.024917	0.009555	0.003129	0.857405	0.00943
7 R	0.602618	0.07708	0.03916	0.013284	0.002123	0.25477	0.010964
7 E	0.040362	0.052046	0.024424	0.009185	0.002434	0.860684	0.010865
8 R	0.628246	0.071532	0.03731	0.012283	0.001529	0.235995	0.013106
8 E	0.038795	0.048185	0.02326	0.008517	0.001793	0.866945	0.012505
9 R	0.681275	0.060909	0.032732	0.010335	0.00096	0.196984	0.016804
9 E	0.036065	0.041294	0.020651	0.007296	0.001192	0.878813	0.014689
10 R	0.780468	0.039005	0.021533	0.006422	0.000419	0.127072	0.025081
10 E	0.031019	0.02794	0.014547	0.004901	0.000601	0.902518	0.018475

Appendix B — Multistart / Solver Evidence

Primary convergence is insufficient: both starts must close and boundary products must agree within 1e-6

N_T	Combined endpoint status	P closure / residual	P / S optimizer termination	S closure / residual	Branch difference	Branch gate
1	CONVERGED / CLOSED	CLOSED / 1.757e-14	Neither separately recorded	CLOSED / 1.510e-14	1.239e-15	PASS
2	CONVERGED / CLOSED	CLOSED / 1.785e-13	Neither separately recorded	CLOSED / 3.642e-14	7.940e-14	PASS
3	CONVERGED / CLOSED	CLOSED / 1.609e-13	Neither separately recorded	CLOSED / 5.596e-14	3.732e-13	PASS
4	CONVERGED / CLOSED	CLOSED / 3.197e-14	Neither separately recorded	CLOSED / 7.816e-14	2.185e-15	PASS
5	CONVERGED / CLOSED	CLOSED / 4.971e-13	Neither separately recorded	CLOSED / 4.885e-14	2.278e-12	PASS
6	CONVERGED / CLOSED	CLOSED / 3.067e-13	Neither separately recorded	CLOSED / 4.974e-14	1.612e-12	PASS
7	CONVERGED / CLOSED	CLOSED / 1.315e-13	Neither separately recorded	CLOSED / 7.017e-14	1.816e-14	PASS
8	CONVERGED / CLOSED	CLOSED / 7.550e-14	Neither separately recorded	CLOSED / 6.928e-14	4.498e-15	PASS
9	CONVERGED / CLOSED	CLOSED / 1.252e-13	Neither separately recorded	CLOSED / 3.375e-14	3.183e-13	PASS
10	CONVERGED / CLOSED	CLOSED / 8.615e-14	Neither separately recorded	CLOSED / 7.017e-14	2.049e-14	PASS

MULTISTART INTERPRETATION BOUNDARY

Matrix 7C CONVERGED/UNCLOSED and closure are combined endpoint statuses, NOT optimizer termination. CLOSED proves both endpoints closed; UNCLOSED does not identify which endpoint failed. Individual optimizer terminations are not separately recorded. Branch gate uses persisted branchReproduced when available. Missing evidence is NOT failure; branch reproduction alone does not establish numerical/stage acceptance. No gate is rerun.

Appendix C — Frozen qualification dispositions

Exact codes where recorded; native persisted flags otherwise. No invented rejection codes.

N_T	Closure	Exact codes / persisted disposition
1	CLOSED	Exact codes not recorded. Persisted accepted: no; numerical acceptance: true; targets: false
2	CLOSED	Exact codes not recorded. Persisted accepted: no; numerical acceptance: true; targets: false
3	CLOSED	Exact codes not recorded. Persisted accepted: no; numerical acceptance: true; targets: false
4	CLOSED	Exact codes not recorded. Persisted accepted: yes; numerical acceptance: true; targets: true
5	CLOSED	Exact codes not recorded. Persisted accepted: yes; numerical acceptance: true; targets: true
6	CLOSED	Exact codes not recorded. Persisted accepted: yes; numerical acceptance: true; targets: true
7	CLOSED	Exact codes not recorded. Persisted accepted: yes; numerical acceptance: true; targets: true
8	CLOSED	Exact codes not recorded. Persisted accepted: yes; numerical acceptance: true; targets: true
9	CLOSED	Exact codes not recorded. Persisted accepted: no; numerical acceptance: false; targets: true; stage 8 acceptance: false; raffinate TPD refinements: false (ROUTINE_REFINEMENT_AMBIGUOUS). TRIAL_NOT_PHYSICALLY_NUMERICALLY_VALID
10	CLOSED	Exact codes not recorded. Persisted accepted: yes; numerical acceptance: true; targets: true

Appendix D — Supplementary global TPD evidence

Legacy supplementary evidence, separate from native matrix per-stage TPD evidence

SUPPLEMENTARY GLOBAL EVIDENCE NOT ATTACHED

This supplementary schema is not attached to the frozen job. Native per-stage stability evidence remains in Appendix A. Absence of this legacy attachment is not a new qualification failure. No current evidence has been substituted.

CANDIDATE SUPPLEMENTARY EVIDENCE NOT ATTACHED

No candidate-generated legacy attachment is recorded. This report does not manufacture it or reclassify the saved matrix result.

Appendix E — Governance / Report Status

7. Governance / Report Status — full immutable provenance is Appendix F

PERSISTED REPORT STATUS — CALIBRATION NOT REQUIRED

Thermodynamic classification	PRE_PILOT_MULTISTAGE_PREDICTIVE_MODEL
Full immutable digests	See provenance appendix
Job ID	145e962e-892a-4424-aab9-91532af9f5ef

Persisted limitations

- Direct thermodynamic sulfur prediction is NOT CALCULABLE. Allocation-based retained sulfur and removal are separate saved estimates.
- The PA representative is non-sulfur-bearing and is not a sulfur surrogate.
- Pilot validation is not established for this persisted snapshot.
- This persisted snapshot is not release eligible.
- SAT is boundary-mole loss; NMP extraction percentage is not applicable.

DOCUMENT CONTROL

Generated automatically and exclusively from the persisted completed job snapshot. No live design inputs were read and no scientific calculation was rerun.

Appendix F — Governance and Provenance

Immutable identifiers and document-control evidence for the completed snapshot

Thermodynamic classification	PRE_PILOT_MULTISTAGE_PREDICTIVE_MODEL
Model SHA-256	f165f0d23eeae810e40341f1cc86a2b97514bc21b74838d4a08eab1760e6e1af
Engine SHA-256	ff35a410554e3f30bb43715691752fdc2f6a016a4689c885bf91272a17c7b486
Input SHA-256	—
Stage-1 authority SHA-256	6ccf5381167d4deb01d313881b63f5bb537c8bc729e3902b2919fc1f4397fbe9
Six-component basis SHA-256	7683442c0419659d83320cfa4433fe8cf298f285f596c4676cf39bb5042838fc
Six-component binding SHA-256	d71161802f9294719416b209739f7918c27cda87a716162fe9308d948271049a
Job ID	145e962e-892a-4424-aab9-91532af9f5ef
Completed at	2026-09-15T07:34:35.795Z

DOCUMENT CONTROL

Generated exclusively from the persisted completed job snapshot. The renderer does not read live Stage-1 or workspace values and does not invoke COSMO-SAC, cascade, TPD, or multistart calculations.

Appendix F — Model and molecular provenance

Only frozen fields or artifacts whose bytes match frozen SHA-256 digests

Input snapshot SHA-256 (report-derived)	278489671d42953a748351174cf76cc30872b4ed5ef3267b4a06a6f868e32907
Result snapshot SHA-256 (report-derived)	22356b58bc8989a6d2be48e4c2396d06f3fabe44b9548abb857165c6ad6c4be7
Digest serialization basis	UTF-8 JSON.stringify of the frozen snapshot objects in stored property order; not a replacement for engine-owned canonical hashes.
cCOSMO native binary SHA-256	a90b34a97bfc9b3fe06ac247a3feae7263f83aa1a29b4d79fea758bec587ce61
Scientific model contract	7C-1.5.0_UNCHANGED
Native runtime versions	{"numpy":"2.1.3","scipy":"1.14.1","python":"3.12.12 (main, Oct 9 2025, 11:07:00) [GCC 15.2.0]","vendorDirectory":"server/research/ecr-pre-pilot-cosmosac/vendor/python","cCOSMOBinarySha256":"a90b34a97bfc9b3fe06ac247a3feae7263f83aa1a29b4d79fea758bec587ce61"}
interaction Form	sum_[pairs,k=0..2] [a+b*(Tref/T-1)]*xi*xj*(xi-xj)^k
fallback Phase Model	null
task238Model Sha256	cb826e81e11143c667dc745677179df76bde0107bcc676d0d52a62078113e1ba
task238Results Sha256	a014064f61e37c8030af34c1893ef1c25c747ccd6750d53147b23f6dcb4955ec
profile Sha256By Family	{"DI":"7e8664f594afa70a232f845dfa9bfb1efd2e10e13c586187b556102a97503ed","PA":"474736e63fd99749f7dad0cd5569959e9dd47a8a2dfe55b6ef80642ed3d1c03e","H2O":"82fc9cc17f71b4b8d28db01b7a25453ae03bea66b4af8adc21d4b85615cd7bb1","NMP":"58dcecc755994f7955aec100dfb26de62c3ad933dcfd25c11f68ddec3b1efa69","SAT":"6f51a75fbfa70df9b410fae88614e5c27d003eea228bc91318edb2854b5c26c2","MONO":"0afc9a2a69c0f3c7827ecdd7553f7ebf1a6e35f8b64597ffc8a6b780c324c275","POLY":"682828484416f124e3207f246d23954197502cba83d9b4744ace239d91fe8854"}
task238Protocol Sha256	7ea9fdc6e893af039a9b1786d1c154b3e9d9cc09bd2aa9fba66d8d023a5e039d
isolated Complist Sha256	ce310440998304063a952323df84b98e4bccc2352a292679bcb70c920fc591a0
active Thermodynamic Model	NATIVE_SEVEN_COMPONENT_CCOSMO_2010_PLUS_ADDITIVE_REDlich_KISTER

Appendix F — Provenance continued

frozen Residual Model Sha256	8bce122869260513e695090bb33478465c43222eb90b6541e47072389241b4f1
frozen Residual Results Sha256	412ea41742d9373f6a3807a188a8cb4d54cbce8e26e61a839a41ed72cf42af29
h2o Generation Manifest Sha256	943b753db28b910e1b672f74571769ec09a781009deccc88db151b64fb780eba
six Generation Manifest Sha256	65baa44c36ffae817e52f60a32591f3fc1be7c140517c8b37c95458c834d1fa9
h2o Profile Verification Sha256	a60f687e1ef953f036c095281bd2461ca212193c35f1d6558e59104441070bf5
task238Parameter Vector Sha256	851fc110959f6ac5622ee94bf04fa6b4396532753f774c33d59e5741570b3c08
historical Residual Model Sha256	8bce122869260513e695090bb33478465c43222eb90b6541e47072389241b4f1
frozen Six Component Runner Sha256	2ab508b252435dd61cea8c601c8bbf4ba0456847de6f5dad73004bf4299732a6
historical Residual Results Sha256	412ea41742d9373f6a3807a188a8cb4d54cbce8e26e61a839a41ed72cf42af29
frozen Six Component Provenance Sha256	3476498d39f96b9761495f11a9e93e626d42aed2eda9f9082fa0a0e29bdd2f5e
frozen Residual Parameter Vector Sha256	fa2227aa99c8bd134525304e68b8f8be5b9fb4b79427fdc520db7920df82827d
inherited Six Component Residual Applied	false
seven Component Runtime Compatibility Sha256	0129f65b9c1c63604dd01bd8516b91e7b742deb7e2cd15b344a2388f24c22e1a
historical Residual Bearing Qualification Runner Sha256	6918807ad2426803b8ea162eb02f0e32a902776323c75e960cacb5296e2e428e
sixGenerationManifestSha256	65baa44c36ffae817e52f60a32591f3fc1be7c140517c8b37c95458c834d1fa9
Hash-verified source	server/research/ecr-pre-pilot-six-component-thermodynamics/generated/generation-manifest.json

Appendix F — Provenance continued

Molecular generation route	RDKit ETKDGv3/MMFF94s lowest conformer; xTB 6.7.1 GFN2-xTB tight gas geometry; CPCM-X 1.1.0 xTB conductor surface
Profile conversion	{"averaging":"NIST Hsieh","benchmarkExecuted":false,"decision":"PROFILE_SE MANTICS_GATE_PASSED","fDecay":3.57,"profileVerificationSha256":"d07ea132252d1a56549629c630b658dcc4d9538e44ca6767f6b9b1fbd15a0e9","rAvSquare dSquareAngstrom":2.3077466748324826,"signMultiplierNativeXtbToTmNist":-1}
h2oGenerationManifestSha256	943b753db28b910e1b672f74571769ec09a781009deccc88db151b64fb780eba
Hash-verified source	server/research/ecr-pre-pilot-seven-component-h2o-profile/generated/generation-manifest.json
Molecular generation route	RDKit ETKDGv3/MMFF94s; pinned xTB 6.7.1 GFN2 tight; pinned CPCM-X 1.1.0; native-xTB sign conversion; NIST-Hsieh sigma3
Profile conversion	{"averaging":"NIST Hsieh","fDecay":3.57,"partitionOrder":["NHB","OH","OT"],"profileVerificationSha256":"a60f687e1ef953f036c095281bd2461ca212193c35f1d6558e59104441070bf5","rAvSquaredSquareAngstrom":2.3077466748324826,"signMultiplierNativeXtbToTmNist":-1}
SAT molecular identity	N-DODECANE; CAS 112-40-3; InChIKey SNRUBQQJIBEYMU-UHFFFAOYSA-N
SAT sigma profile SHA-256	6f51a75fbfa70df9b410fae88614e5c27d003eea228bc91318edb2854b5c26c2
SAT optimized geometry SHA-256	5e6932017e292e36b6fedc08d49c063723c2f22a6c89a7cd498118a4b99e2780
SAT surface SHA-256	81bd1e4ddba823a3276c4aef0841b10446181183877ac77889c0215b44400fdf
MONO molecular identity	N-PROPYLBENZENE; CAS 103-65-1; InChIKey ODLMAHJVESYWTB-UHFFFAOYSA-N
MONO sigma profile SHA-256	0afc9a2a69c0f3c7827ecdd7553f7ebf1a6e35f8b64597ffc8a6b780c324c275
MONO optimized geometry SHA-256	c9c920c752fc4ea0dae8156d222035afdfed555462e73a2d71eaaf770d1262e3
MONO surface SHA-256	77733c6aec2cb649d1272c13f0325d3c433db36da59834a95e7754f2379e5057

Appendix F — Provenance continued

DI molecular identity	1-METHYLNAPHTHALENE; CAS 90-12-0; InChIKey QPUYECUOLPXSR-UHFFFAOYSA-N
DI sigma profile SHA-256	7e8664f594afa70a232f845dfafbfbb1efd2e10e13c586187b556102a97503ed
DI optimized geometry SHA-256	a06e943ab6c44586469a1d7aee09e69b5bc4f0cfc617f62ed1c42e1e0faf5b7a
DI surface SHA-256	21233caaf6d73dad8c3b1531c8e86860ae92f07c0bd9297ab0ba7b46c325e33
POLY molecular identity	PYRENE; CAS 129-00-0; InChIKey BBAAQIROQSPTKN-UHFFFAOYSA-N
POLY sigma profile SHA-256	682828484416f124e3207f246d23954197502cba83d9b4744ace239d91fe8854
POLY optimized geometry SHA-256	43982af9f0d9094e97cecc443acdd879d8d9c345ff49b6fd02f6a86fb19903796
POLY surface SHA-256	ee2fa575ae0daef7513cc14e414c5de6eb8ba08633dc3f851ab544a45d67b232
PA molecular identity	BIS-P-CUMYLPHENYL-AMINE; CAS 10081-67-1; InChIKey UJAWGGOCYUPCPS-UHFFFAOYSA-N
PA sigma profile SHA-256	474736e63fd99749f7dad0cd5569959e9dd47a8a2dfe55b6ef80642ed3d1c03e
PA optimized geometry SHA-256	18297ad328e27487a186808d9bcb5cbc660ffd3595250f05c83c19a3d8c971dd
PA surface SHA-256	3022c951ebb62a131d2eb969de7ba4956078c2ec3200c2bc0fa4ba4b3f43467e
NMP molecular identity	N-METHYL-2-PYRROLIDONE; CAS 872-50-4; InChIKey SECXISVLQFMRJM-UHFFFAOYSA-N
NMP sigma profile SHA-256	58dcecc755994f7955aec100dfb26de62c3ad933dcfd25c11f68dddec3b1efa69
NMP optimized geometry SHA-256	ec8ff8189bda773187773b174e982cc012e800266022aa3197b4019c923ffb6
NMP surface SHA-256	837958143169284b19e2678ad65d6a39349a059c80057b300df3e1868d761d76

Appendix F — Provenance continued

H2O molecular identity	WATER; CAS 7732-18-5; InChIKey XLYOFNOQVPJJNP-UHFFFAOYSA-N
H2O sigma profile SHA-256	82fc9cc17f71b4b8d28db01b7a25453ae03bea66b4af8adc21d4b85615cd7bb1
H2O optimized geometry SHA-256	52e69af7d53aff3f3d673d7aaf00829da2332cf1c2ba9d77780194ed05d930dd
H2O surface SHA-256	2c4a345271076f9a812f1813422be7152f015b687f6fd1d51968f6e381f5c043